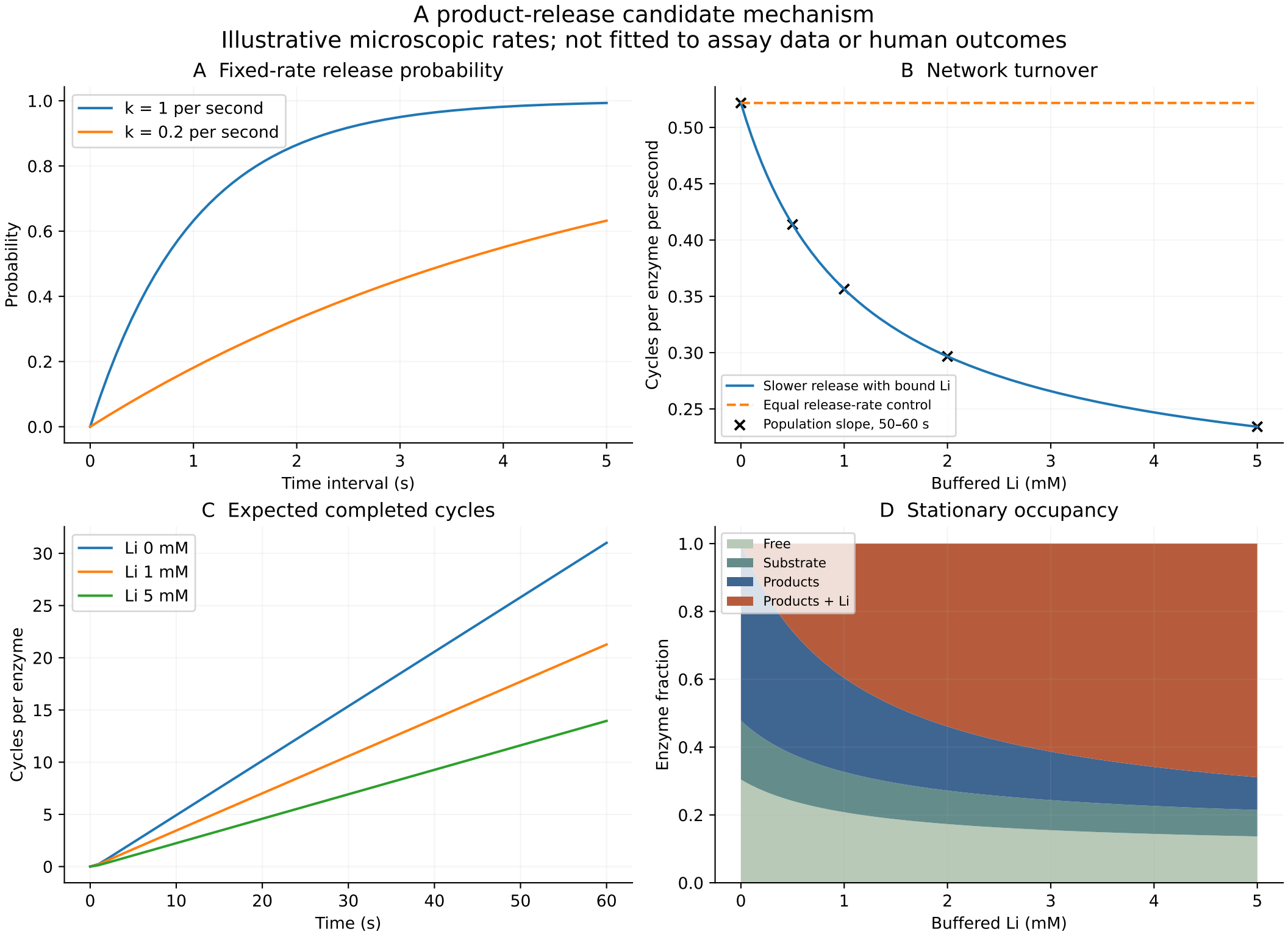
A product-release candidate mechanism
Illustrative microscopic rates; not fitted to assay data or human outcomes
A Fixed-rate release probability
k = 1 per second
k = 0.2 per second
Probability
Time interval (s)
B Network turnover
Slower release with bound Li
Equal release-rate control
Population slope, 50–60 s
Cycles per enzyme per second
Buffered Li (mM)
C Expected completed cycles
Li 0 mM
Li 1 mM
Li 5 mM
Cycles per enzyme
Time (s)
D Stationary occupancy
Free
Substrate
Products
Products + Li
Enzyme fraction
Buffered Li (mM)